SPICE Simulation

Test your circuit before you solder

KiCad ngspice • LTspice • Models • Analyses

Audience: Beginner → Intermediate Engineering Students

DC • AC • Transient • Noise • Monte Carlo • Convergence

Why Simulate?

Faster, cheaper, safer than building

Speed

Try 10 designs in an hour
vs 10 weeks on the bench.

Safety

No fires, no smoke, no
broken parts.

Insight

See node voltages everywhere,
not just where probes reach.

Optimisation

Sweep parameters,
find sweet spot.

Documentation

Simulated proof attached
to design reviews.

Learning

Build intuition faster
than real-world experiments.

What is SPICE?

The grandfather of electronic simulators

- SPICE = Simulation Program with Integrated Circuit Emphasis
- Born 1971 at UC Berkeley — academic free tool
- Solves circuit equations using Newton-Raphson + matrix methods
- Modern variants: ngspice (open-source), LTspice (free, Analog Devices), HSPICE (commercial)
- KiCad embeds ngspice — simulate directly from the schematic
- PSpice / TINA / Multisim — commercial alternatives
- Industry uses SPICE for analog circuits universally
- Some digital simulation possible — but not the main strength

Tool Comparison

Pick the right SPICE

Tool	Cost	Pros	Cons
LTspice	Free	Fast, stable, huge model library	Schematic UI dated
KiCad + ngspice	Free / OSS	Integrated with your schematic	Slower convergence
TINA-TI	Free	TI-supported, op-amp focus	TI parts only
PSpice (Cadence)	₹2-5 L/yr	Industry standard	Expensive
Multisim	₹50k+	Educational, easy	Limited models
Qucs-S	Free	Modern UI, RF/AC focus	Smaller community

Recommendation

Start with LTspice — fast, free, ten thousand examples online. Move to KiCad+ngspice when you want simulation tied to your real schematic.

Analysis Types

What questions SPICE can answer

DC operating point

Steady-state voltages
and currents.

DC sweep

Sweep a source —
plot transfer characteristic.

AC analysis

Small-signal frequency
response (Bode plot).

Transient

Time-domain behaviour —
oscillators, switching.

Noise

Output noise vs frequency.
Good for analog frontends.

Monte Carlo

Vary components randomly —
yield analysis.

SPICE Models

What makes simulation accurate

- Every component needs a model — set of equations describing behaviour
- Passive R/L/C: built-in primitives
- Semiconductor: vendor-supplied (.lib, .mod, .sub files)
- Op-amps: 'macro models' — simplified equivalent circuits
- MOSFETs: BSIM level-3, level-7, level-49 — different fidelity
- Where to get models: vendor websites, LTspice library, IBIS models for digital
- Convergence problems often come from low-quality models — try simpler ones
- Validate models: simulate a known circuit, compare to datasheet

KiCad Simulation Workflow

From schematic to plot

- Build the circuit in Eeschema — use SIM-compatible symbols
- Assign SPICE models to each part (Edit Symbol → SPICE Model)
- Add sources: V1 (DC), VAC (AC), VSIN (sinusoidal), VPULSE (transient)
- Open Tools → SPICE Simulator (or in KiCad 8+: integrated sim panel)
- Configure analysis type (DC/AC/Trans)
- Pick signals to probe (right-click net → 'Probe')
- Run simulation → results in a plot window
- Iterate: change values, re-run, learn

Useful Circuits to Simulate

Practice projects for students

Voltage divider

DC sweep, see how output varies with input.

RC filter

AC sweep, find -3 dB point and roll-off slope.

Op-amp inverter

Trans + AC: bandwidth, slew rate.

LC oscillator

Trans: see startup, frequency, harmonics.

Buck regulator

Trans + worst-case Vin: ripple, efficiency, control loop.

MOSFET driver

Trans with gate cap: rise time, switching loss.

Convergence Problems

When SPICE refuses to solve

- 'Time step too small' — try larger maxstep or different integration method
- 'Singular matrix' — likely two voltage sources in parallel or zero-resistance loop
- Add tiny resistors (1 mΩ) in parallel paths to prevent singularities
- Add small caps to ground at every node — helps DC convergence
- Use .ic (initial condition) statements to give SPICE a starting point
- Replace ideal sources with high-impedance equivalents
- Reduce simulation step / increase tolerance with .OPTIONS
- Simplify the model — start crude, add detail incrementally

Parameter Sweeps & Optimisation

Find the best design automatically

- .STEP statement sweeps a parameter (R1, C1, temp, etc.)
- Plot multiple curves to compare designs at a glance
- .MEAS extracts numbers (bandwidth, gain, rise-time)
- Output to .csv, plot in Python / Excel for deeper analysis
- Monte Carlo (.STEP MC) varies components randomly
- Yield analysis: % of designs meeting spec
- Optimisation: feedback loop adjusts components automatically (Qucs-S, ngspice)

Common SPICE Mistakes

What trips up beginners

- Missing GND node (node 0) — SPICE requires a reference
- Two voltage sources directly in parallel (or two C in series with no path)
- Wrong model for the part (using BJT model for FET)
- Step too coarse — transient misses the fast edge
- Forgetting to set the right operating point — clipping at saturation
- Trust the simulation more than the bench — models have limits
- Ignoring parasitics — real PCBs have inductance, capacitance
- Simulating ideal switches but expecting real-world ringing

Beyond SPICE

When you need more than circuit simulation

MATLAB / Simulink

Control system, DSP,
closed-loop simulation.

Python (NumPy)

Custom scripts, data
analysis, modelling.

Verilog / VHDL

Digital logic +
FPGA / ASIC simulation.

Cadence ADS

RF + microwave system
simulation.

COMSOL

Multiphysics — thermal,
magnetics, electromagnetics.

KiCad SI tools

Future: signal integrity
for high-speed digital.

Best Practices

Habits of a skilled simulator

- Start crude, refine iteratively
- Validate model on a known circuit
- Add parasitics for realistic results
- Use real datasheet numbers, not text-book values
- Add small ESR / ESL to caps & inductors
- Capture screenshots for design reviews
- Re-run after each component change
- Sweep parameters to find sensitivity
- Monte Carlo for production yield
- Always cross-check with bench measurement
- Save schematic + simulation file in repo
- Document the 'why' — not just the 'what'

Key Takeaways

Key takeaways from this lecture

LTspice = best free tool

Start here.

KiCad embeds ngspice

Simulate from real schematic.

Models = accuracy

Vendor SPICE models matter.

Multiple analysis types

DC, AC, Transient, Noise.

Sweeps find optima

Don't pick by gut feel.

Validate on bench

Sim builds intuition — bench confirms.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- LTspice — free download from Analog Devices
- KiCad SPICE tutorials (docs.kicad.org)
- Robert Cordell — 'Designing Audio Power Amplifier Circuits' (LTspice examples)
- Gilbert Sirois — LTspice videos (Analog Devices)
- Phil's Lab YouTube — practical SPICE walkthroughs
- Vendor SPICE models: TI, ADI, Microchip, Infineon
- ngspice manual (free)
- EEVblog forum simulation threads

Thank you • Build something this week.